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Soft Switching in Switched Inertance Hydraulic Circuits

Switched inertance hydraulic systems (SIHS) use inductive, capacitive, and switching elements to boost or "buck" (reduce) a pressure from a source to a load in an ideally lossless manner. Real SIHS circuits suffer a variety of energy losses, with throttling of flow during transitions of the high-speed valve resulting in as much as 44% of overall losses. These throttling energy losses can be mitigated by applying the analog of zero-voltage-switching, a soft switching strategy, adopted from power electronics. In the soft switching circuit, the flow that would otherwise be throttled across the transitioning valve is stored in a capacitive element and bypassed through check valves in parallel with the switching valves. To evaluate the effectiveness of soft switching in a boost converter SIHS, a lumped parameter model was constructed. Simulation demonstrates that soft switching improves the efficiency of the modeled circuit by 42% at peak load power and extends the power delivery capabilities by 77%. [DOI: 10.1115/1.4036887]

Introduction

Switched inertance hydraulic systems (SIHS) are switch mode fluid power circuits analogous to buck and boost voltage converters in the field of power electronics. Buck and boost converters provide an ideally lossless method for converting an input direct current (DC) voltage to an output DC voltage. Converter circuits produce a DC output voltage that may be lower than the input voltage in the case of a buck converter or greater than to input voltage in the case of a boost converter [1]. Hydraulic buck and boost converters behave as DC–DC pressure transformers and can be applied in place of throttling valve control or mechanical-hydraulic pressure boosting transformers in circuits where multiple actuators are powered from a common, constant pressure rail.

Fluid power SIHS circuits were proposed by Brown et al. as an efficient alternative to lossy throttling valve control [2]. SIHS circuits utilize fluid mass in a long conduit as an inertive device that has the same function as the inductor in the power electronic circuit.

In application, switch mode fluid power circuits experience pipe flow resistance losses and orifice throttling losses during valve transitions. Rannow and Li identified that valve transitions accounted for 60% of the overall losses in the virtually variable displacement pump configuration that they simulated [3]. They discussed mitigating these losses by placing a small spring piston accumulator in parallel with the tank side switching valve. This piston accumulator was a passive device through the first half of the piston travel. The second half of the piston travel was restricted by an active locking mechanism which released the piston at the onset of the tank valve opening transition. This locking mechanism was critical in the function of the proposed soft switching method. A pressure-based locking piston accumulator for this purpose was designed by Van de Ven that eliminated the need for external control over the locking mechanism [4]. This design was experimentally validated by Beckstrand and Van de Ven [5]. Beckstrand and Van de Ven found that the devices performed as expected under a narrow range of experimental conditions, but the implementation of complicated geometry of the locking device posed a challenge to the device over wider operating ranges.

Power electronic converters also experience switching and conduction losses [6]. To mitigate switching losses, "zero voltage switching" techniques are applied. This approach seeks to eliminate the voltage drop, and hence losses, across a transitioning switch. Henze et al. proposed adding capacitors and diodes in parallel with the transistors in a boost converter to achieve zero voltage switching [6]. In his circuit, a switch overlap is implemented, during which time the circuit enters a resonant phase due to the inductance in the circuit and the added capacitors. The resonance of the circuit charges or discharges the capacitors to a voltage that is equal to either the load or low-supply voltage. Diodes in parallel with the capacitors prevent voltage over or undershoot during this resonant phase.

The Hydraulic Boost Converter. In this work, the hydraulic boost converter, depicted in Fig. 1, is used as a basis for efficiency improvement. This circuit is referred to as the "baseline" circuit in contrast to the soft switched circuit presented later.

The duty cycle of the boost converter is the ratio of time that flow is directed to tank over the switching period. An ideal boost converter behaves according to the following relationship:

$$p_{\text{load}} = \frac{p_{\text{rail}}}{1 - D} \tag{1}$$

where p_{load} is the load pressure, p_{rail} is the supply pressure, and D is the duty cycle. Similarly, and by conservation of power, a

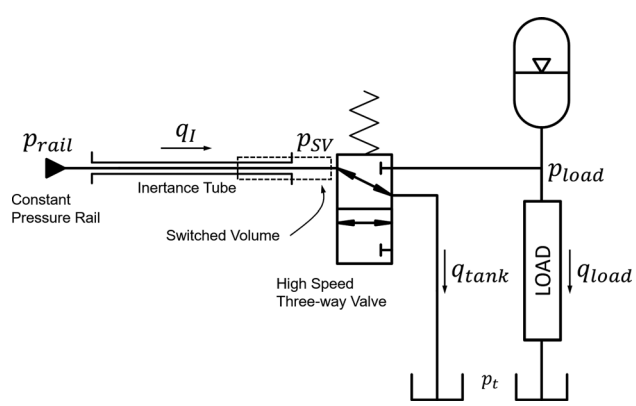

Fig. 1 Baseline boost converter hydraulic circuit

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lossless boost converter would have the following relationship between mean rail flow rate and load flow rate:

$$q_{\text{rail}} = \frac{q_{\text{load}}}{1 - D} \tag{2}$$

where q_{load} is the mean load flow rate and q_{rail} is the mean rail flow rate.

Work conducted by Wang et al. explains baseline boost converter operation and presents simulation results for a boost converter operating over a range of load flow conditions [7]. De Negri et al. describe load pressure behavior given a linear inertance tube resistance model with analytical equations [8]. These models do not capture valve losses but agreed well with experimental results when an oversized and relatively fast transitioning valve was used. Pan et al. developed a frequency domain model of the entire boost converter system that captured inertance tube resistance but assumed linear valve orifice resistance and instantaneous transitions [9]. These models were modified in a following paper, in which the inertance tube was modeled in the frequency domain and the rest of the system was modeled in the time domain [10]. This mixed time–frequency approach, which was utilized by Manhartsgruber et al. [11], allows simulation of wave delay in the inertance tube as well as simulation of nonlinear behavior such as orifice resistance and variable orifice area in the rest of the system.

In this work, the inertance tube is modeled utilizing a lumped parameter time domain model, which is valid for conduit lengths less than 4% of the switching wavelength [12,13]. These models offer a simplified but valid look at system performance in the absence of the complex transients introduced by wave delay effects for boost converters with short inertance tubes. The purpose of this work is to gain fundamental perspective on the interaction of various passive circuit elements, which the complicating effects of a distributed parameter pipeline model would obscure. A high-speed valve model, based on a crank-slider-actuated valve capable of 100 Hz switching and 0.8 ms transition, is included to represent real valve losses [14].

In the Soft Switching section, a soft switching hydraulic boost converter is proposed to mitigate valve transition losses. Description of the circuit modeling procedure is next, followed by description of the simulation procedure. Numerical results are presented in the Simulation Results section, with discussion following. The final section makes concluding remarks.

Soft Switching

Soft switching in the context of this work is a switching methodology rather than a particular device. The concepts herein are adapted from the zero-voltage switching methodology applied to electronic boost converters by Henze et al. [6]. The goal of soft switching is to reduce switching valve transition losses by mitigating pressure gradients across the valve during periods of transition.

Theory of Operation. An ideal switch mode circuit with instantaneous valve transition and lossless fluid flow can modulate power in absence of losses. However, a physical system experiences finite valve transition time, which results in forcing flow through a partially opened orifice. These throttling events result in a large pressure drop, per the orifice equation. Power losses due to throttling flow across an orifice are expressed in the following equation:

$$P_{\text{loss}} = \Delta p q = \Delta p C_d A \sqrt{\frac{2\Delta p}{\rho}} \tag{3}$$

where A is the instantaneous orifice area. From Eq. (3), if the pressure differential across the orifice can be reduced to zero, throttling losses are eliminated.

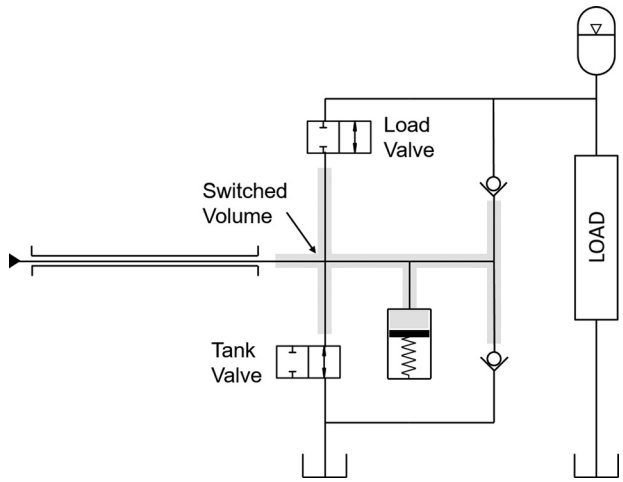

Fig. 2 Soft switching boost converter utilizing spring loaded capacitive elements

A soft switched boost converter has an additional spring loaded accumulator and check valves applied in parallel with the switching valves, as shown in Fig. 2. These passive elements modulate the pressure in the "switched volume" at the outlet of the inertance tube, adjacent to the switching valves. The pressure differential, and hence power loss across a transitioning valve, is minimized by charging this volume to the load pressure or discharging this volume to the tank pressure during a valve overlap period prior to those valve opening events. The switched volume is charged by a positive flow rate through the inertance tube during valve overlap and is discharged by negative flow rate through the inertance tube during valve overlap in the following switching sequence. The capacitive element also buffers the pressure in the switched volume against sudden change during valve closing events. The three-way valve in Fig. 1 is separated into two two-way valves in Fig. 2 for clarity of discussion.

Sequence of Operation:

Figure 3(*a*)	The cycle is initialized by flowing to tank, building velocity and thus kinetic energy and momentum in the fluid in the inertance tube. The load accumulator is large enough that it can supply flow to the load for the duration of time that the pressure supply is shunted to tank.
Figure 3(*b*)	At time DT, the tank valve is closed. The momentum of the flow in the inertance tube charges the switched volume accumulator (SVA). Once the pressure in the switched volume reaches the load pressure, flow is allowed through the load check valve, equalizing the pressure across the load side valve.
Figure 3(*c*)	The load side valve is opened across a minimal pressure differential. The adverse pressure gradient causes the flow in the inertance tube to decelerate and eventually reverse.
Figure 3(*d*)	The flow direction in the inertance tube is now towards toward the pressure supply. The load side valve closes and the momentum of the flow decompresses the switched volume. Once the switched volume reaches tank pressure, the tank side check valve opens, equalizing pressure across the tank side valve. The tank side valve now opens across a minimal pressure differential, initiating the next switching cycle.

Calculating Capacitance for Soft Switching. The soft switching boost converter is a direct hydraulic analogy of the zero

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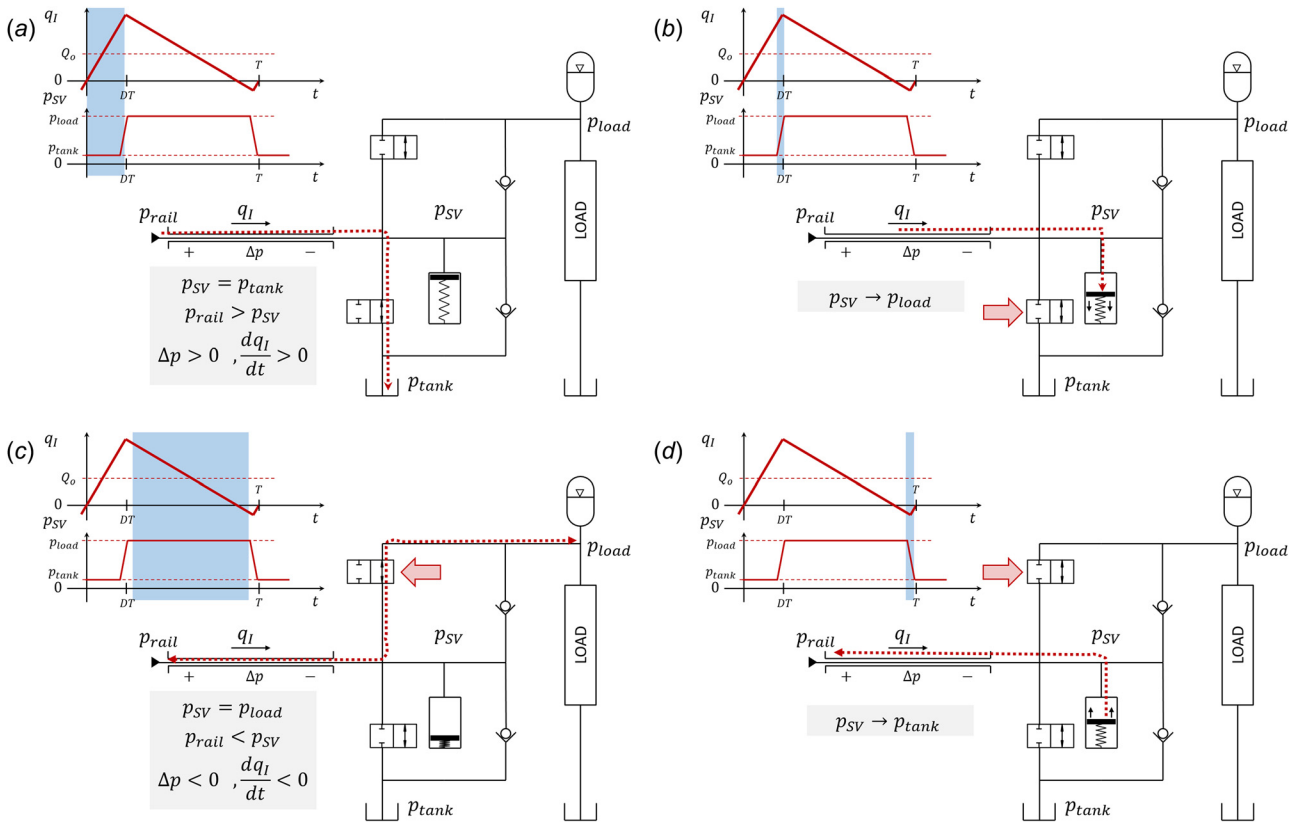

Fig. 3 Soft switched boost converter sequence of operation

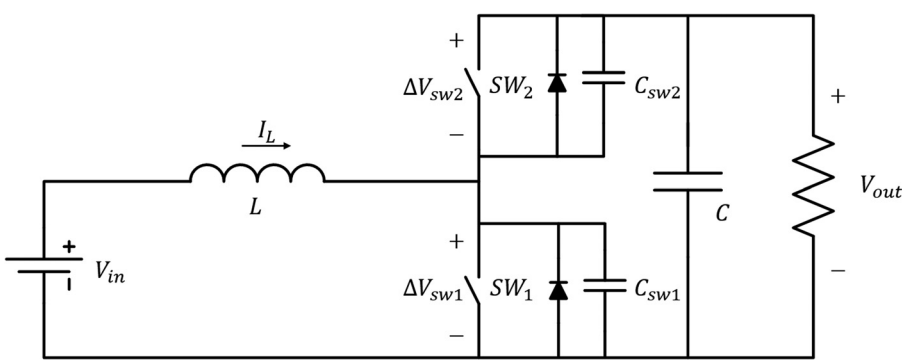

Fig. 4 Zero voltage switching boost converter. Image adapted from Ref. [6].

voltage resonant transition circuit topology proposed by Henze et al. and shown in Fig. 4 [6]. The lumped parameter modeling hydraulic circuit approach pursued in this work enables direct analogies to be made with the electronic circuit. Henze et al. present several analytical relationships that are used to size the capacitive elements with respect to electrical current flow and switch overlap time and provide a useful starting point to sizing the hydraulic components.

When both switches are open, the soft switched boost converter in Fig. 4 reduces to a resonant inductive–capacitive (L–C) circuit. Henze et al. propose the relationship in Eq. (4) to ensure that the capacitors are fully charged during this period

$$T_{\mathrm{OL}} > \frac{2C_{\mathrm{sw}}\Delta V_{\mathrm{sw}}}{|I_L|} \tag{4}$$

where T_{OL} is the length of time that both switches are open, C_{sw} is the capacitance of one of the capacitors in parallel with the switch, ΔV_{sw} is the voltage differential across the switch, and I_L is the current in the inductor at the time of the switching event. It is assumed that T_{OL} is small relative to the switching period T and that the switch transition time is small relative to both T_{OL} and T. An inequality appears in Eq. (4) due to the parallel diode which allows current to bypass the capacitor once fully charged or discharged. The limiting condition for determining the minimum switching overlap time is the electric current level at the SW_1 switch opening transition. At this time, the current is at a low-magnitude negative value, a similar state as the hydraulic boost converter in Fig. 3(*d*).

Power electronics have the benefit of short transition times, often several orders of magnitude shorter than the switching period. To minimize the switching period in hydraulic switch mode circuits, the transition time of the valve is on the order of 5% of the switching period. Due to the relatively long transition period of the hydraulic valve, a substantial capacitance is required to buffer the switched volume pressure during the transition.

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Equation (4) assumes negligible discharge of the capacitance during the transition period and does not establish a minimum capacitive value. Assuming a linear switching valve orifice area, it is reasonable to add 50% of the transition time to the overlap time in Eq. (4) to account for capacitor discharge during the relatively long switch transition time. With this considered, a capacitance value that will buffer the switched volume during transition but also fully charge during the valve overlap can be calculated directly. Restating in hydraulic terms, Eq. (4) becomes

$$T_{\mathrm{OL}} + \frac{1}{2}T_{\mathrm{TR}} = \frac{C_{\mathrm{SV}}\Delta p}{q_I} \tag{5}$$

where T_{TR} is the valve transition time, C_{SV} is the hydraulic capacitance of the switched volume, Δp is the pressure differential between the supply and tank when calculating the tank side transition or supply and load when calculating the load side transition, and q_I is the inertance tube flow rate at the time of transition.

Equations (6) and (7) show hydraulic capacitance values for a spring-piston accumulator and a compressible fluid volume, respectively,

$$C = \frac{A_p^2}{k_s} \tag{6}$$

$$C = \frac{V_c}{\beta} \tag{7}$$

where C is the hydraulic capacitance, A_p is the area of the piston, k_s is the spring rate, V_c is the volume of the compressible fluid, and β is the bulk modulus of the fluid.

The tank closing transition event in Fig. 3(*b*) takes place at peak flow, and the load closing transition event in Fig. 3(*d*) at minimum inertance tube flow. At moderate duty cycles, the pressure differential Δp term can be considered approximately equal prior to load side or tank side transitions. Assuming equal transition time and overlap time, the ideal switched volume capacitance for each of these transition events is proportional to the flow rate magnitude at the time of the transition event according Eq. (5). This means that the capacitive value of the SVA will be a compromise between optimal switching to load or to tank, lying somewhere between the two values predicted by Eq. (5) when either peak or minimum flow is considered.

To address the asymmetry in capacitance required for effective switching at the high-flow and low-flow transition events, the piston travel can be limited such that the capacitance of the accumulator goes to zero above a fixed pressure. At switched volume pressures greater than this cutoff pressure, further piston travel is restricted and the capacitance of the switched volume is only due to the compliance of the fluid in the inertance tube and the SVA cylinder, per Eq. (7). Doing this allows a large capacitance at the low-pressure/high-flow conditions that correspond to the tank valve transition and smaller capacitance at the high-pressure/low-flow conditions that occur at load valve transition.

Model

Inertance Tube Model. Ideally, the inertance tube is purely an inertive element. However, viscous flow through a conduit results in friction losses and compliance of the fluid within the conduit results in capacitive effects. Both of these nonideal terms are distributed spatially throughout a physical system. The physical system experiences wave delay in the transmission of pressure signals from one end of the conduit to the other, which travel at the sonic velocity. In this work, a lumped parameter approximation is applied to the inertance tube in order to better illustrate the soft switching concept by simplifying the complex system dynamics that occur in wave delay models. To maintain validity of the lumped parameter modeling approach, the inertance tube length is set to 4% of the switching period wavelength, calculated as

$$\lambda = cT \tag{8}$$

where λ is the switching period wavelength and c is the sonic velocity in the fluid. Lumped parameter systems convey information from one computational node to the next instantaneously, but application of the inertance tube length, $l_{\mathrm{tube}} = 0.04\lambda$, standard minimizes the effect of wave delay to the point that it can be accurately neglected.

The "medium transmission line" modeling approximation is adapted from the field of power electronics to represent the inertive, resistive, and capacitive elements of the inertance tube [13]. The inertance tube in Fig. 5 is modeled as series inertive and resistive elements, with parallel capacitive elements at either end. Each capacitive element represents the compliance of one half of the total fluid volume contained in the tube.

In Eq. (9), L is the inertance of the fluid in the inertance tube, calculated as

$$L = \frac{\rho l_{\mathrm{tube}}}{A} \tag{9}$$

where ρ is the density of the hydraulic fluid and A is the cross-sectional area of the inertance tube. R is the hydraulic resistance of the tube, where in the laminar case

$$R = \frac{8\pi\mu l_{\mathrm{tube}}}{A_{\mathrm{tube}}^2} \tag{10}$$

where μ is the kinematic viscosity of the hydraulic fluid. Laminar friction was taken as a starting point in the circuit model. Simulation later demonstrated that the flow in the inertance tube did remain in the laminar regime.

C_1 and C_2 are the capacitance values of the oil volumes at either end of the inertance tube model

$$C_1 = C_2 = \frac{\frac{1}{2}V_{\mathrm{tube}}}{\beta} \tag{11}$$

where V_{tube} is the volume of the inertance tube and β is the bulk modulus of the hydraulic fluid.

The flow rate in the inertive element is a function of the inertance in the tube, and the pressure difference applied across the tube, less the pressure drop due to resistive losses

$$p_{\mathrm{in}} - p_{\mathrm{out}} = L\frac{dq_I}{dt} + q_I R \tag{12}$$

where p_{in} is the inertance tube inlet pressure, p_{out} is the outlet pressure, and q_I is the flow rate through the inertive lumped element.

The pressures in the inlet and outlet volumes are given by the following equations:

$$\frac{dp_{\mathrm{in}}}{dt} = \frac{1}{C_1}(q_{\mathrm{in}} - q_I) \tag{13}$$

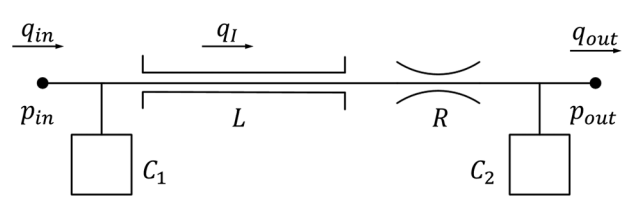

Fig. 5 Lumped parameter inertance tube "medium line" model

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$$\frac{dp_{\text{out}}}{dt} = \frac{1}{C_2}(q_I - q_{\text{out}}) \tag{14}$$

In the boost converter simulation, the inertance tube inlet is connected to the rail pressure, which is modeled as a constant pressure boundary condition. A pressure source applied to p_{in} in Fig. 5 eliminates the need to model capacitance C_1.

Switching Valve Model. The modeled boost converter is driven by a simulated crank-slider three-way valve switching at 100 Hz [14]. This valve model is simplified, approximating the load and tank sides as separate valves with port orifices of 3×10^{-5} m^2. The modeled valve area profile is linear and transitions in 0.8 ms. The switching valve is assumed to have zero leakage.

The flow rates through the load and tank branches of the three-way valve are calculated by the orifice equation

$$q_v = C_d A_v(t)\sqrt{\frac{2\Delta p_v}{\rho}} \tag{15}$$

where q_v is the flow rate through the valve, C_d is the discharge coefficient, $A_v(t)$ is the instantaneous valve open area, and Δp_v is the pressure differential across the valve.

Load Model. The load on the circuit is modeled as a constant volumetric flow rate sink.

Check Valve Model. The check valves are modeled as ideal elements that instantaneously open once the cracking pressure has been exceeded. Once the check valve is open, it is modeled as an orifice

$$q_c = \begin{cases} 0 & \text{if } p_1 - p_2 < p_{\text{crack}} \\ C_d A_c \sqrt{\dfrac{2(p_1 - p_2 - p_{\text{crack}})}{\rho}} & \text{if } p_1 - p_2 > p_{\text{crack}} \end{cases} \tag{16}$$

where q_c is the cracking pressure of the check valve, A_c is the orifice area of the check valve, p_{crack} is the check valve cracking pressure, and p_1 and p_2 are the pressures at the inlet and outlet of the check valve, respectively. The check valve areas in the model are set to 75% of the switching valve orifice area, to ensure that the check valves close once the switching valve has opened. The cracking pressure of both check valves was set to 0.10 MPa.

Gas Charged Load Accumulator Model. The load accumulator is charged with nitrogen and modeled as an ideal gas undergoing isentropic expansion and compression

$$p_{\text{acc}} V_{\text{gas}}^{\gamma} = p_{\text{charge}} V_{\text{acc}}^{\gamma} \tag{17}$$

where p_{acc} is the instantaneous accumulator pressure, p_{charge} is the accumulator charge pressure, V_{acc} is the accumulator volume, γ is the specific heat ratio of the charge gas, and V_{gas} is the instantaneous volume of the charge gas.

Spring-Piston Switched Volume Accumulator Model. The spring-piston accumulator applied at the switched volume is modeled as a purely capacitive device

$$\frac{dp_{\text{SV}}}{dt} = \frac{1}{C_{\text{SVA}}} q_{\text{SVA}} \tag{18}$$

where C_{SVA} is the capacitance of the SVA, per Eq. (6), and q_{SVA} is the flow rate into the spring piston accumulator.

Circuit Efficiency. The efficiency of the boost converter is calculated by dividing the power delivered to the load over a cycle by the power supplied by the rail, averaged over a cycle

$$\eta = \int_t^{t+T} \frac{q_{\text{load}}(\tau)(p_{\text{load}}(\tau) - p_{\text{tank}})}{q_I(\tau) p_{\text{rail}}} d\tau \tag{19}$$

Simulation Results

The simulated boost converter circuits were developed around a 6 MPa rail pressure and load flow rate of 0.1 L/s. The tank pressure is modeled at an elevated pressure of 300 kPa to avoid any instants of negative pressure in the circuit, which would indicate the occurrence of cavitation. In each simulation, the inertance tube is initialized at zero flow and at rail pressure. The load is initialized at ambient pressure and given flow rate. The simulation proceeds until the load pressure achieves steady state for 20 switching cycles, defined as less than 0.1% mean pressure variation between consecutive cycles. Simulated duty cycles are $0.1 \leq D \leq 0.9$. Duty cycles less than 0.1 and greater than 0.9 experience insufficient valve opening for proper operation due to the valve transition time. The hydraulic fluid properties used in the simulation are listed in Table 1. Note that a constant bulk modulus is assumed to simplify the model and maintain focus on illustrating the role of soft switching.

The baseline boost converter was simulated with an inertance tube length of 0.506 m. The variation in inertance tube flow over a cycle is a function of the inertance value of the tube, with longer and thinner tubes having higher inertance and correspondingly lower variation in flow [8]. The inertance tube diameter for the baseline circuit was selected such that continuous positive flow rate through the inertance tube is achieved over the range of duty cycles at 0.1 L/s load flow rate. Ensuring only positive flow rate through the inertance tube prevents the soft switching effect of discharging the switched volume prior to tank valve transition as occurs in Fig. 3(*d*). The De Negri et al. analytical model, which includes laminar pipe friction [8], predicts this value to be 3.3 mm. The simulations presented in this work include switching valve resistance and finite transition time, which are not captured by the De Negri et al. model. The simulations demonstrate that an inertance tube diameter of 3.9 mm results in continuous positive flow in the inertance tube throughout the boost converter operational range when considering valve losses. Figure 6 presents the simulated baseline boost converter inertance tube flow rates over a range of duty cycles for an inertance tube length $l_{\text{tube}} = 0.506$ m and inertance tube diameter $d_{\text{tube}} = 3.9$ mm.

Next, a single switching cycle during cyclic steady-state operation of the baseline boost converter is analyzed. The parameters of the simulated baseline and soft switched boost converters are outlined in Table 2, and the performance of the converters is outlined in Table 3. Full open valve losses are the power losses that the flow would experience when passing through a fully opened valve. Transition losses are the valve losses in excess of the full open valve losses. Friction losses are the laminar flow losses in the inertance tube.

The baseline boost converter experiences pressure spikes at $t = 0.35$ s and $t = 0.65$ s in Fig. 7. The pressure spikes occur when the flow in the inertance tube is forced through the transitioning valve orifices. These spikes result in strong adverse pressure gradients in the inertance tube, which decelerate the inertance tube flow at the same temporal locations.

Table 1 Physical properties of the simulated hydraulic fluid

Hydraulic fluid properties	
Density (kg/m^3)	874
Bulk modulus (GPa)	1.40
Kinematic viscosity (Pa · s)	0.0404
Entrained air fraction (%)	0

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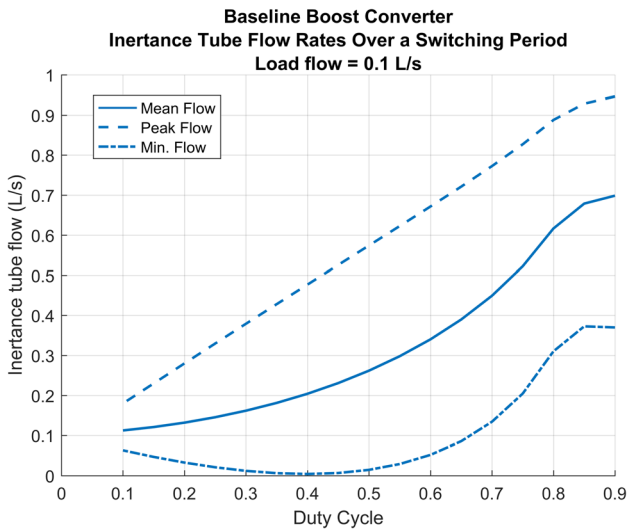

Fig. 6 Simulation results of inertance tube mean, peak and minimum flow rates over a switching period versus duty cycle for the baseline boost converter

Table 2 Physical parameters of the simulated boost converters

Boost converter simulation parameters		
	Baseline	Soft switched
Switching speed (Hz)	100	
Valve overlap (ms)	0.3	
Duty	0.7	
Valve orifice area (m^2)	3×10^{-5}	
Valve transition time (ms)	0.8	
Inertance tube length (m)	0.506	
Inertance tube diameter (mm)	3.9	
Load flow rate (L/s)	0.1	
Switched volume accumulator:		
Piston diameter (mm)	N/A	3.6
Piston travel (mm)	N/A	3.0
Spring rate (N/m)	N/A	10,000

Table 3 Baseline boost converter performance metrics

Boost converter performance, $D = 0.7$		
	Baseline	Soft switched
Efficiency, η (%)	29.71	60.51
Load pressure (MPa)	7.97	11.65
Transition losses (J/cycle)	8.13	1.39
Full open valve losses (J/cycle)	1.85	0.70
Friction losses (J/cycle)	8.62	5.45

The losses over a cycle are shown in Fig. 8. The most significant loss event occurs at the load valve opening transition at $0.35\,\mathrm{s} < t < 0.4\,\mathrm{s}$. The major pressure spike that occurs prior to this event results in a large pressure differential across the valve and, hence, major throttling losses.

The parameters of the simulated soft switched system are identical to those of the baseline system with the exception of the added SVA and check valves. The SVA piston hits a stop once the internal pressure reaches 3 MPa, effectively disabling the accumulator until the pressure goes below this level. The soft switched converter exhibits 82% lower transition losses relative to the baseline system, reducing per cycle transition losses from 8.13 J/cycle

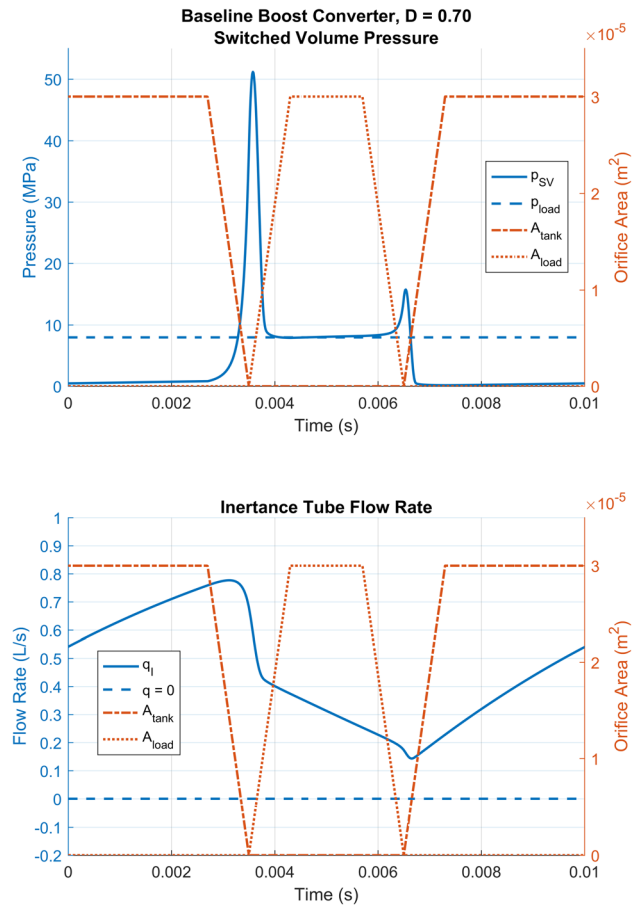

Fig. 7 Switched volume pressure inertance tube flow in a baseline boost converter, $D = 0.7$

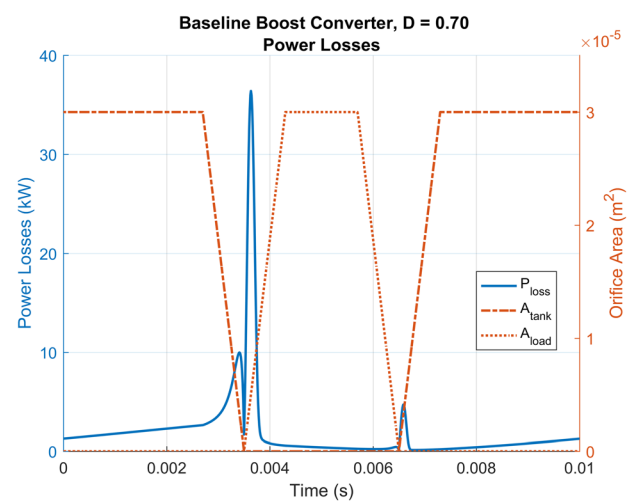

Fig. 8 Power losses over a switching cycle in a baseline boost converter, $D = 0.7$

to 1.39 J/cycle. Figure 9 shows the soft switched system dynamics, on the same scales as Fig. 7.

In the soft switched boost converter, the SVA absorbs flow during the tank closing event, slowing the rate at which the pressure rises in the switched volume. At time $t = 0.0029$ s in Fig. 9, the piston hits the bottom of its travel, negating its capacitive function. After this time, the pressure ramps up quickly as a result of the much lower capacitive value of the inertance tube fluid compliance. The pressure spike prior to the load valve opening event

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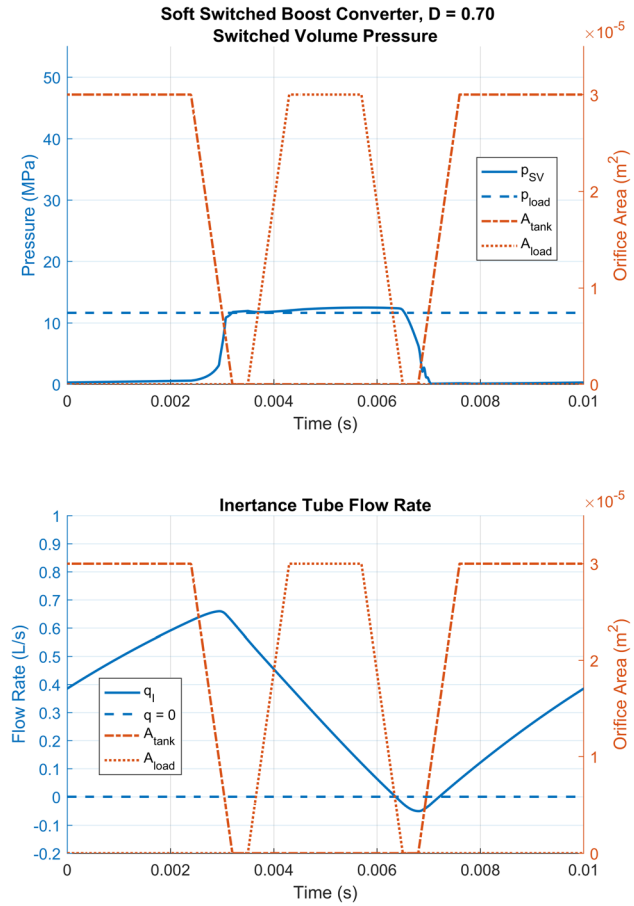

Fig. 9 Switched volume pressure and inertance tube flow in a soft switched boost converter, $D = 0.7$

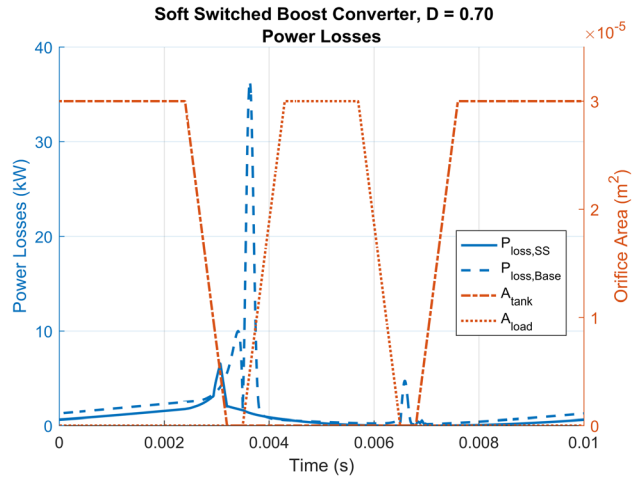

Fig. 10 Power losses over a switching cycle in a soft switched boost converter compared to power losses in a baseline boost converter, $D = 0.7$

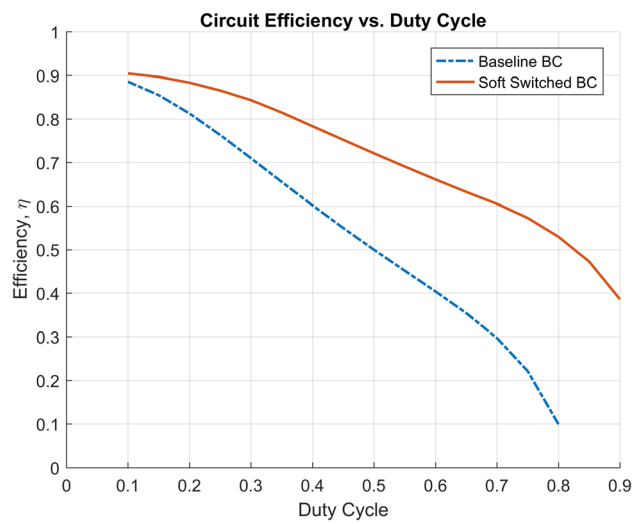

Fig. 11 Efficiency versus duty cycle for a baseline and soft switched circuit. Load flow rate is 0.1 L/s.

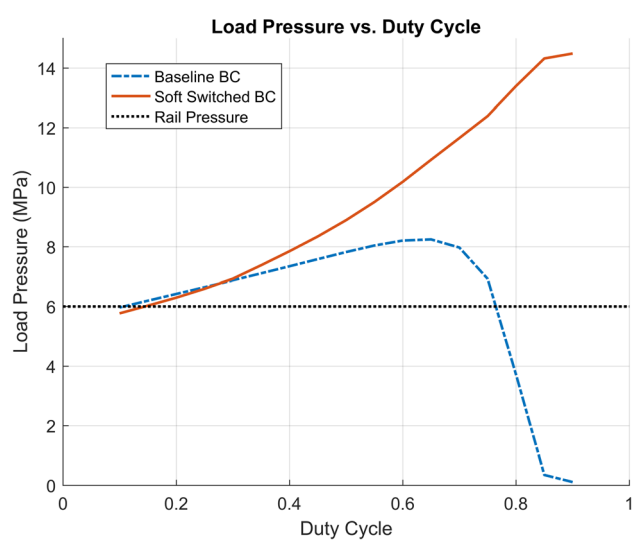

Fig. 12 Load pressure versus duty cycle for a baseline and soft switched circuit. Load flow rate is 0.1 L/s.

is mitigated by the check valve in parallel with the load valve, which provides an alternate flow path once the switched volume pressure is greater than the load pressure. The pressure in the switched volume falls during the overlap after the load valve closes. At time $t = 0.0068$ s, the accumulator piston begins traveling, discharging fully at time $t = 0.007$ s.

Comparing the flow rate plots in Figs. 7 and 9, the mean flow in the soft switched boost converter inertance tube is lower than the mean flow rate in the baseline boost converter. This is due to the baseline circuit having greater flow through the inertance tube than that predicted by Eq. (2). This excess flow, termed "flow loss," is related to the efficiency of the circuit. Less efficient circuits experience greater flow loss at the same duty cycle and load flow rate [9]. The increased mean flow rate through the inertance tube of the baseline boost converter results in greater friction and fully open valve losses relative to the soft switched boost converter.

The power loss of the soft switched boost converter over a cycle is compared to that of the baseline converter in Fig. 10. In this figure, it is clear that the major transition loss events are eliminated or reduced in the soft switching circuit.

Figures 7–10 focus on the cyclic steady-state circuit behavior at a duty cycle of 0.7. Figure 11 compares the efficiency of the two systems over a range of duty cycles.

The efficiency results for the baseline boost converter are similar to the results predicted and experimentally verified by De Negri et al. [8]. The soft switched boost converter offers efficiency improvements over the range of operation relative to the baseline circuit, which results in greater load pressure for a given duty than the baseline circuit, as shown in Fig. 12.

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The increased load pressure that the soft switched circuit can deliver results in greater power delivery across the range of duty cycles as shown in Fig. 13. The soft switched boost converter is able to deliver nearly double the maximum power of the baseline boost converter. At duty cycles $D > 0.7$, the load pressure delivered by the baseline circuit drops due to circuit losses. This drop in load pressure delivery is seen where the baseline boost converter load power bends back toward lower power values and lower efficiency.

Discussion

The SVA was designed for a fixed load flow rate of 0.1 L/s, with the boost converter operating at a duty cycle $D = 0.7$. This SVA, combined with the check valves, resulted in performance improvements across the range of operation. The system would likely benefit from an optimization process to select the best SVA capacitance value and cutoff pressure considering the range of duty cycles and various load flow rates.

The SVA presented in the work is small and has a high spring rate. Such a spring-piston accumulator would be difficult to produce. Calculating the required capacitance value for the SVA via Eq. (5) results in values of $7.0 \times 10^{-14}\ \mathrm{m^3/Pa}$ at the high-flow, tank closing transition and $1.2 \times 10^{-14}\ \mathrm{m^3/Pa}$ at the low-flow, load closing transition.

Studies of distributed parameter switch mode converter circuits have found that the length of the inertance tube, switching frequency, and duty cycle should have the following relationships for maximum efficiency [7,9,15]:

$$\begin{aligned} l_{\mathrm{tube}} &= \frac{TDc}{2} \qquad 0 < D < 0.5 \\ l_{\mathrm{tube}} &= \frac{T(1-D)c}{2} \qquad 0.5 \le D < 1 \end{aligned} \tag{20}$$

These formulas result in an optimal inertance tube length of 25% of the maximum switching speed wavelength, where the switching wavelength is calculated by Eq. (8). The 100 Hz switching and sonic velocity of 1265 m/s result in an inertance tube length of 3.16 m. This inertance tube length is outside of the validity of lumped parameter modeling, but the outlet capacitive value of such a tube can be approximated by Eq. (11), given a tube diameter.

An inertance tube of 3.16 m in length and having diameter of 7.5 mm has an outlet capacitive value of $4.9 \times 10^{-14}\ \mathrm{m^3/Pa}$. This indicates that the compliance of the fluid in the inertance tube alone is enough to achieve soft switching, eliminating the need for the SVA when optimal length inertance tubes are applied. The diameter of the inertance tube affects the inertive, capacitive, and resistive values of the conduit, all of which have a bearing on circuit performance. For this reason, the best inertance tube diameter would be discovered through optimization.

The check valves that allow flow to bypass the switching valves prior to valve opening events must operate quickly relative to the switching period. Ideally, these valves operate instantaneously, but a physical system will rely on valves that open in a fraction of a millisecond. Research into small high-speed plate type hydraulic check valves has demonstrated them to be effective in a 300 Hz pumping application with potential to operate at 400–500 Hz [16]. Reed style check valves with low moving mass have demonstrated high-speed potential in automotive applications and are currently being studied for hydraulic applications [17].

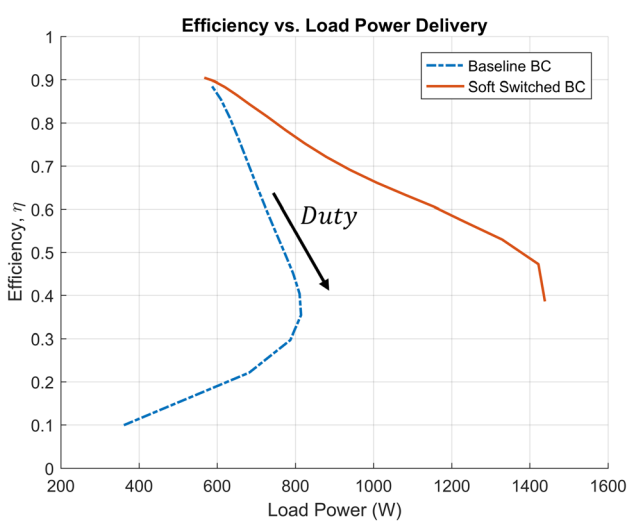

Fig. 13 Efficiency versus load power delivery for a baseline and soft switched circuit. Load flow rate is 0.1 L/s.

Conclusion

A soft switching boost converter circuit was presented as a direct hydraulic analogy to the soft switching boost converter in the field of power electronics. This circuit aims to reduce the losses due to flow throttling through a transitioning switching valve. The circuit adds check valves in parallel with the switching valves and a spring piston accumulator at the outlet of the inertance tube. All elements added to the circuit are passive and do not require external control for operation. The nonsoft switched and soft switched boost converters were simulated using a lumped parameter approach. The lumped parameter models remain valid as the length of the inertance tube was restricted to 4% of the switching wavelength, minimizing wave delay effects.

The results of the soft switched boost converter were compared to a boost converter of equal inertance tube dimensions lacking soft switching strategies. The modeled soft switching boost converter provided up to 42% gains in efficiency and 77% improvement in load power delivery for a fixed load flow rate of 0.1 L/s. When the length of the inertance tube is increased close to an optimal length, the inherent capacitance of the fluid in the inertance tube is sufficient to achieve soft switching, eliminating the need for an accumulator in the switched volume. Distributed parameter models are required to capture the physics in optimally sized inertance tubes and should be applied in future work.

This work focuses on soft switching in a boost converter. Henze et al. present a similar soft switching buck converter that could readily be adapted to a hydraulic circuit using the approach presented in this paper. The soft switching buck converter also relies on passive elements without the need for external, mechanical control.

Nomenclature

A = orifice area, $\mathrm{m^2}$
A_p = piston area, $\mathrm{m^2}$
A_{tube} = cross-sectional area of the inertance tube, $\mathrm{m^2}$
c = sonic velocity, $\mathrm{m/s}$
C_d = orifice discharge coefficient
C_{SV} = hydraulic capacitance of the switched volume, $\mathrm{m^3/Pa}$
C_{SVA} = hydraulic capacitance of the SVA
C_{sw} = capacitance in parallel with switch, F
D = duty cycle, with respect to tank valve open time
d_{tube} = inertance tube diameter, m
I_L = current through inductor at time of switching, A
k_s = spring rate, $\mathrm{N/m}$
L = hydraulic inertance of the fluid in the inertance tube, $\mathrm{kg/m^4}$
l_{tube} = inertance tube length, m
p_{acc} = accumulator pressure, Pa
p_{charge} = accumulator precharge pressure, Pa
p_{in} = inertance tube inlet pressure, Pa
p_{load} = mean load pressure over a switching cycle, Pa
P_{loss} = power loss due to orifice throttling, W

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p_{out} = inertance tube outlet pressure, Pa
p_{rail} = supply pressure, Pa
p_{SV} = switched volume pressure, Pa
p_{tank} = circuit return pressure, Pa
p_1 = check valve upstream pressure, Pa
p_2 = check valve downstream pressure, Pa
q = volumetric flow rate across orifice, m^3/s
q_c = check valve volumetric flow rate
q_I = volumetric flow rate into switched volume at time of transition, m^3/s
q_{load} = mean load flow rate, m^3/s
q_{rail} = mean rail flow rate, m^3/s
q_{SVA} = volumetric flow rate into the SVA, m^3/s
R = hydraulic resistance the inertance tube due to pipe friction, $Pa/(m^3/s)$
SVA = switched volume accumulator
T = switching period, s
T_{OL} = valve or switch overlap, s
T_{TR} = valve transition time, s
V_{acc} = accumulator volume, m^3
V_c = compressible volume, m^3
V_{gas} = gas volume inside accumulator, m^3
V_{tube} = inertance tube volume, m^3
β = bulk modulus, Pa
Δp = pressure drop across orifice, Pa
ΔV_{sw} = voltage across switch, V
η = circuit hydraulic efficiency
λ = switching period wavelength, m
ρ = working fluid mass density, kg/m^3

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